Marketing ROI & Budget Reallocation Strategy

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Goal, KPIs & Attribution Rule

Business Goal

- Optimize marketing budget allocation to maximize revenue and improve channel efficiency.

North Star KPI

- Total Attributed Revenue

Supporting KPIs

- ROAS (Return on Ad Spend)
- CAC (Customer Acquisition Cost)
- Conversion Rate (CVR)
- Total Orders

Attribution Rule Used

- Last-touch attribution model:
  Revenue is assigned to the final marketing channel that directly drove conversion.

Data & Methodology

Data Sources

- fact_channel_daily
- fact_campaign_data
- fact_sessions

ETL & Data Preparation

- Cleaned null and blank values
- Removed revenue outliers using capped revenue
- Built star schema model
- Created calculated measures (ROAS, CAC, CVR)

Key Metric Definitions

ROAS = Revenue ÷ Spend
CAC = Spend ÷ Orders
Conversion Rate = Orders ÷ Sessions
Regression Impact = Spend × performance-weighted proxy

Key Insights

- Organic channel delivers the highest ROAS (~12.8) with relatively low spend.
- Email also shows strong efficiency (~5.0 ROAS).
- Paid Social has the lowest ROAS (~0.64) despite high investment.
- Search drives the largest total revenue but moderate efficiency (~1.3 ROAS).
- Returning users generate the majority of revenue.
- Web performs better than mobile across most channels.
- Budget allocation is currently concentrated in lower-efficiency channels.

Attribution Findings

What Looks Good

•Organic and Email channels are highly efficient.
•Search contributes significant revenue volume.

What Looks Weak

•Paid Social is overspending relative to revenue generated.
•Referral channel shows moderate performance but limited scalability.

Interpretation

Last-touch attribution favors bottom-funnel channels and may over-credit conversion-focused traffic while undervaluing assist channels.

Regression Findings

Estimated Incremental Impact

Regression-based proxy suggests:

Some channels may be over-attributed under last-touch logic.

Incremental impact varies when controlling for overall performance.

Key Differences

- Attribution assigns direct credit.
- Regression estimates true incremental contribution.

Limitations

- Proxy regression (not full MMM model).
- No seasonality controls.
- Limited time horizon.
- No impression-level data.

Risks & Next Steps

Risks

- Attribution bias toward last-touch channels
- Potential channel saturation
- Data limitations
- Short-term performance volatility

Next Steps

- Conduct controlled budget reallocation experiment
- Implement multi-touch attribution model
- Collect impression and customer journey data
- Build full marketing mix regression model
- Monitor 30-day incremental lift